

Touch My Heart: Navigating the Heart Models in MR with Haptic Feedback, 3D Sound, and Interactive Gamification

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ABSTRACT

Touch My Heart utilizes a mixed reality headset and digital twin technology for an immersive exploration of the heart. The system, integrating hand tracking and haptic feedback, offers a realistic simulation, enhancing spatial audio for a fusion of senses. Users can interact with a 3D heart model, providing a comprehensive understanding of structure, function, and abnormalities including cardiac arrhythmia's and valvular heart diseases. The initiative supports medical education, enabling detailed exploration of heart pathophysiology and collaborative discussions among healthcare professionals. Additionally, it facilitates patient engagement by educating on heart conditions.

Index Terms: Human-centered computing—Visualization—Visualization design and evaluation methods

1 INTRODUCTION

Mixed Reality (MR) transforms the boundary between physical and digital realms, particularly in healthcare where it reshapes medical education, diagnostic visualization, and patient engagement. A notable application is the visualization of the human heart, exemplified by *Touch My Heart* as a standalone Unity app anchored to a haptic device for an enhanced immersive experience [8]. This iteration introduces features for medical education and gamification, incorporating authentic heart dynamics through measurements with a digital stethoscope [1]. The heart model, validated by a cardiologist, offers interactions with three cardiac conditions, detailed ECGs, and synchronized spatial sound recordings. Rigorous verification ensures medical correctness, with the entire setup visualized in figure 1.

This demonstration marks a significant evolution from the initial *Touch My Heart* project, progressing beyond a simplified heart model responding to wristwatch heart rates [9]. The current version introduces cross-sectional visualization, 3D sound, refined tactile feedback, disease representations, and a gamification module. It represents a major stride in providing a more immersive and educational cardiac visualization experience.

Touch My Heart serves diverse purposes today, including collaboration, medical education, and patient engagement, with future potential in personalized healthcare and expanding to cover various organs and medical conditions.

2 SYSTEM ARCHITECTURE

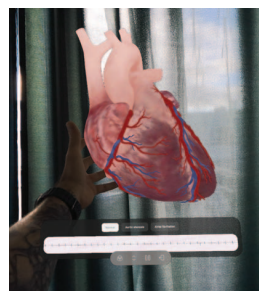
The central hardware component is the Magic Leap 2 augmented reality headset, serving as the primary interface for users to interact with the virtual environment. The Magic Leap 2 device is connected to a personal computer (PC) via a USB-C cable, primarily for device charging and optional device streaming if a WiFi connection is

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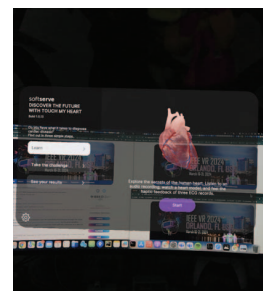
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Figure 1: Visualization of the *Touch My Heart* setup.



(a) HeartModel



(b) MainMenu

Figure 2: Application view

unavailable. This ensures a consistent power supply to the headset, maintaining an uninterrupted user experience. The figure 2 illustrates the system view and figure 3 its architecture respectively.

The UltraHaptics device, a pivotal element in enhancing the sense of touch in mid-air, is also an integral part of the system architecture. The UltraHaptics device is connected to the PC through two USB-A ports. The first USB-A port is dedicated to the UltraLeap camera, responsible for accurate hand tracking, while the second port facilitates the functioning of the ultrasonic part of the device, providing natural tactile feedback.

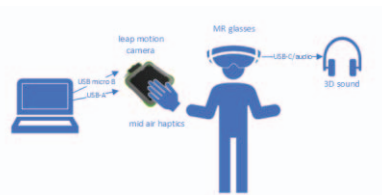


Figure 3: System setup.

3 HEART CASES

In a healthy physiological state, the heart comprises four chambers – two atria (left and right) separated by the atrial septum, and two ventricles (left and right) separated by the intraventricular septum. The right ventricle pumps deoxygenated blood to the pulmonary artery, while the left ventricle pumps oxygen-rich blood to the aorta. Electro-mechanical coupling ensures synchronized contractions, maintaining regular heart function.

Atrial fibrillation (AF), the most common heart rhythm disorder, affects 2-4% of the population, with a rising prevalence predicted to reach 14-17 million individuals in the European Union by 2030 [4], [2]. AF increases the risk of stroke, myocardial infarction, and heart failure. Clinically, it manifests as irregular and rapid heart rhythms, potentially causing symptoms like palpitations and shortness of breath. Diagnosis relies on physical examination and ECG.

Aortic stenosis (AS) is the predominant primary valve disease in Europe and North America, affecting approximately 12.5% of those aged 75 and older, with an estimated global population of 13 million individuals with AS. Despite its progressive nature, AS can be asymptomatic until severe blood flow restriction occurs. Auscultation, particularly a high-pitched, crescendo-decrescendo murmur in the second right intercostal space, plays a pivotal role in diagnosis. Untreated AS is associated with poor prognosis, emphasizing the importance of early detection and intervention.

4 DYNAMIC OF A HEART MODEL

To attain the required quality and representation, a medically precise 3D model of the human heart was employed. To generate diverse animations depicting various heart conditions, a procedural approach that involved translating authentic heart audio signals into animation curves was adopted. This method provided the essential flexibility for refining the final outcome. The ultimate animated output was then fused with the original audio to achieve a believable and accurate representation.

Procedural modelling: The task was to adjust a heart model for precise animation of different heart diseases. Using a non-destructive procedural method, both realism and adaptability for frequent refinements were achieved. That opened the possibility to dive into the realm of medical visualization and advanced 3D modeling, suitable for enthusiasts and professionals alike.

Shading and lighting in Unity: As the concluding phase, the procedural shader graphs were crafted to establish the ultimate appearance and texture for a heart model designed for real-time applications. An intriguing challenge was to enable interactive slicing of the heart, functioning seamlessly across various axes.

5 HAPTIC FEEDBACK

Ultrasound mid-air haptics [3, 7] is a novel technology that uses ultrasound speakers to induce a vibrotactile sensation on the user's skin, specifically with the use of ultrasonic wave interference phenomenon [5]. This enables a person not to hear the ultrasound, but to feel its touch, generated by the multiple speakers interfering together. This results in the sensation of 'touch without touching' or the virtual touch [6]. In this study, the tactile feedback was generated by an Ultraleap STRATOS Explore (USX) device composed of a 16x16 ultrasonic transducer array [10]. The haptic pattern presented to the participant's palm is a circle with a dynamically changing radius oscillating between 2 and 5 cm creating a sensation of a heart beating.

6 GAMIFICATION

Users can test themselves and try to define the heart diseases in a form of a game. The challenge may be passed either with registration (done on user's mobile phone) or without registration (they go through the challenge but will not appear in the leaderboard). The person who got the highest score - had the most correct answers

and was the fastest could win a prize. When launching the challenge mode, user will be prompted to answer three questions. In each case the person has to select the type of heart rate out of three options based on visual, haptic and sound experience. Questions are randomised, so we never know what is the order, and which type of heart rate popups in which position. If the answer is false – 0 points are given. If the answer is correct, the points are calculated basing on the time the user spent for the particular question – the quicker the answer the higher the points. (999 is the max). When the challenge finished the total score appears, registered users will see themselves in the leaderboard with their corresponding name and result.

7 3D SOUND

Spatial audio cues align with heart structure spatial relationships, allowing users to discern static sound object directions and establish connections with visual elements. The user's interaction capabilities include navigation through position and orientation adjustments and manipulation of sound objects within the model.

8 SUMMARY

The *Touch My Heart* demo takes heart exploration to the next level. It combines haptic feedback and heart holograms with spatial audio for a truly immersive experience. This demo is perfect for students and patients alike, offering a clear understanding of heart function, disorders, and their treatment implications without the need for invasive procedures. It's an exciting way to peer inside the human body's beating heart. We invite you to a discussion about its usage in the healthcare, educational spheres, and beyond.

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